

SMART INDIA HACKATHON 2026



AM-CORD AI:

Autonomous Mobility Coordination AI

- **Problem Statement ID** – SIH26123
- **Problem Statement Title**- Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots (AMRs) in Smart Warehouses.
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- 168016
- **Team Name** : Array of Excellence

Prototype:

- **Demo Video link:** <https://drive.google.com/drive/folders/14oOErbjm0gizr23FtAVYCs8Qg8JZI2ah?usp=sharing>
- **Github repo link** : <https://github.com/adityaadept2008/SIH>
- **Dashboard UI link:** <https://smart-warehouse-mu.vercel.app>





PROPOSED SOLUTION

- **Decentralized Multi-Robot Coordination:** Multiple Autonomous Mobile Robots (AMRs) collaboratively execute warehouse tasks without relying on a permanent central controller, ensuring scalability and fault tolerance
- **AI-Powered Dynamic Route Intelligence:** Machine Learning predicts corridor congestion, travel time, and waiting delays from live fleet telemetry, enabling robots to proactively select faster and less congested routes.
- **Learning-Guided Task Allocation:** CBBA assigns warehouse tasks using predicted completion time instead of only shortest distance, resulting in balanced workload distribution and improved fleet efficiency.
- **Vision-Based Warehouse Awareness :** Deep Learning detects pallets, cartons, and temporary aisle blockages using onboard cameras, providing additional environmental intelligence for dynamic rerouting.
- **Adaptive Route Optimization:** Reinforcement Learning continuously learns from warehouse operations to recommend better routing strategies that minimize congestion and improve long-term fleet throughput.
- **Deterministic Safety-First Architecture:** CO-WHCA* trajectory reservations, corridor leases, ORCA and the LiDAR-based Safety Supervisor verify every AI recommendation before execution.
- **Continuous Learning from Operations:** The system automatically generates simulation datasets from robot telemetry, enabling periodic retraining so fleet intelligence continuously improves without manual data collection.

ML based ETA prediction

DL based safety object
detection system

INNOVATION &
UNIQUENESS

Continuous
Improvement using RL

Hybrid approach(ML+
WHCA* deterministic)

```
[ 106.4s ] * [CBBA DONE ] Task rnd_task_002
[ 106.4s ] ▶ [TASK ACCEPT ] robot_1 navigatin
[ 155.8s ] ⚡ [PICKUP ARRIVED] robot_1 reached
[ 155.9s ] → [PICKUP DONE ] robot_1 finished
[ 206.4s ] ⚡ [DROPOFF ARRIVED] robot_1 reached
[ 206.5s ] ✓ [TASK COMPLETED] rnd_task_002 fini
```

How Concepts Work

If there's no central controller,
how do we coordinate?

Robots make decisions locally and share state through
Zenoh-assisted ROS 2 communication. A router improves
startup discovery without becoming a fleet decision controller.

How do we actually decide
which robot gets a task?

CBBA works like an auction. Each robot calculates a
completion cost, then the fleet agrees on one winner
before assigning the task.

How do we make sure our
planned routes don't clash?

ML estimates travel time and congestion. CO-WHCA* then
verifies the route against short time-space reservations so robots
avoid conflicting paths.

What if two of us need the same
narrow aisle? How do we avoid
collision if someone entered our path?

One AMR receives an exclusive lease for a narrow aisle. ORCA
adjusts motion in real time, while LiDAR safety can slow or stop
the robot.



TECHNICAL APPROACH

Frontend Arch.

1. Fleet Overview Dashboard
2. Live Digital Twin Map
3. Thermal Congestion Heatmap
4. Distributed Mutex Visualizer
5. Task Queue & CBBA Monitor
6. Manual Task Injector
7. Fleet Statistical Analytics
8. Plain-English Activity Feed
9. Telemetry & Audit Exporter

1. Gazebo Harmonic + ROS 2 Jazzy

Physics, Wheel Slip, Namespaced AMRs (/amr_1..4)

2. Zenoh-Assisted ROS 2 Coordination

Router-assisted discovery with Cyclone DDS fallback

3. DL Perception (YOLO Object Detection)

OAK-D YOLOv8 (Box/Pallet/Worker) + 2D LiDAR Grid Marking

4. 2D Topological Grid Engine

0.5m Aisle-Aware Grid A* (Replaces Euclidean)

5. Edge ML ETA Regressor (ONNX)

22-Feature Delay Pred. (>10ms Fallback to Dist/v)

6. CBBA Task Allocation Engine

Decentralized P2P Auctions + Leased Epochs

7. CO-WHCA* Space-Time and Corridor Mutex

Short reservation windows and single-aisle access control

8. ORCA Dynamic Avoidance

Reciprocal 50/50 Velocity Avoidance at Crossings

9. Deterministic Safety Supervisor

ISO 3691-4 TTC < 0.6m → E-Stop (cmd_vel = 0.0)

10. Differential Drive Motor Actuation

Wheel Execution via Verified /cmd_vel

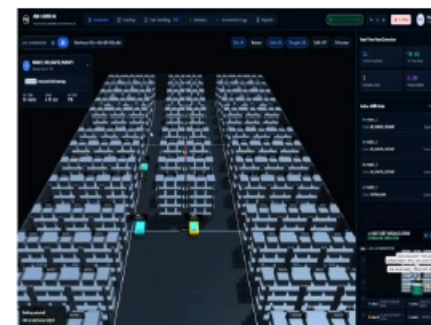
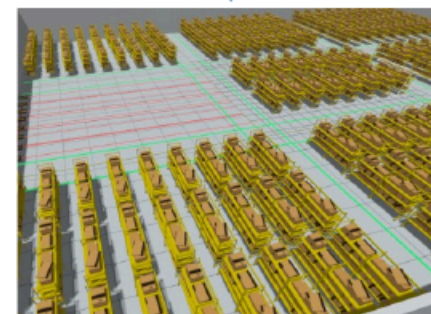
11. Continuous RL Flywheel

12k+ Run Telemetry → MAPPO Background Retraining

12. Mission Control Digital Twin

WebSocket Bridge → Three.js 2D/3D & Heatmaps

Warehouse with all edge conditions for testing.



Tech Stack :

Tools and software: ROS 2 Jazzy, Gazebo Harmonic, Eclipse Zenoh (rmw_zenoh_cpp), Cyclone DDS fallback, WebSockets, Ubuntu 24.04 LTS.

AI roadmap: Python, PyTorch, YOLOv8, LightGBM/XGBoost ETA models and MAPPO reinforcement learning.

Target hardware: Raspberry Pi 4 / Jetson-class edge devices, TurtleBot 4, 2D LiDAR and optional RGB-D camera.

IMPLEMENTATION PROCESS

Physics Simulation & Verified AMR Spawn Gating

DL Perception & 2D Topological Grid Mapping (Replacing Euclidean)

ML-Guided Decentralized Task Allocation (CBBA + Dynamic ETA)

Space-Time Planning (WHCA*) & Distributed Corridor Mutex

Deterministic 4-Tier Safety Gate & Mobile Wi-Fi Mesh

Continuous RL Optimization & Statistical Digital Twin Dashboard



FEASIBILITY AND VIABILITY



SMART INDIA
HACKATHON
2026

Implemented Autonomy

CBBA task allocation, CO-WHCA* time-space planning, trajectory reservations, distributed corridor access, ORCA, LiDAR safety veto, obstacle rerouting, docking, battery simulation and telemetry logging.

Operational Frontend & Easy Control

Live 2D/3D map, robot paths, ETA, speed, battery, task status, AMR inspection, task dispatch and priority, congestion heatmaps, readable decision logs, charging views and operational reports.

Deployment Viability

Open-source, modular and vendor-independent architecture designed for commodity Raspberry Pi/Jetson-class edge hardware, without a permanent central fleet controller.

CHALLENGES

- Warehouse Wi-Fi/AP failure, packet loss, network partitions and added mesh-routing latency
- ML uncertainty, model drift, biased training data and vision misclassification
- Congestion, narrow-aisle deadlocks, temporary obstacles and robot failure during an active task



SOLUTIONS

- ✓ Zenoh supports router-assisted discovery for dense ROS 2 process graphs. Robots keep local planning and safety authority; Cyclone DDS remains a fallback.
- ✓ ML ETA predictions act as a soft cost for CBBA bidding and CO-WHCA* route selection. If confidence is low or inference fails, deterministic planning takes over.
- ✓ Completed missions expand the dataset, retraining improves makespan and waiting time
- ✓ Missing updates are detected and critical messages are automatically re-sent. Robots ignore old or duplicate information; if communication remains unreliable, they slow down and avoid entering shared corridors until fresh data is received.
- ✓ The DL model identifies pallet, carton or person for the dashboard. LiDAR confirms physical occupancy and the local Safety Supervisor retains final braking authority.

IMPACT ON TARGETED AUDIENCE



Warehouse & Logistic Operators

Decentralized autonomy: AMRs retain local planning and safety authority. Zenoh reduces dense-process startup discovery through router registrations; the dashboard is not a control dependency.



E-Commerce Fulfillment Centers

20% Makespan Optimization: Space-time WHCA reservations and corridor mutex eliminates bottleneck congestion



Systems Integrators & Robotics Engineers

Serverless Edge: Runs on Raspberry Pi 4/ Jetson Hardware using open-source ROS-2, needs no expensive cloud software



Floor Supervisors & Technicians

Autonomous Failure Recovery: CBBA monotonic lease expiry stalled tasks in <2s without manual dispatch

BENCHMARKS

0 Inter Robot Collisions
(Hard Sensor Veto)

> 20% Makespan Gain Vs
Stop & Wait Approach

Zenoh-Assisted Discovery
 $O(N)$ router registrations vs. $O(N^2)$ DDS peer discovery

Reallocation Latency <2s
(for Robot Crash)

Decentralized Fault Isolation

Robots use unique IDs, numbered messages, and expanding safety buffers to prevent collisions and confusion when Wi-Fi cuts out.

Multi Tiered Safety Boundary

Smart routing and on-device AI optimize travel times, while hardware-level LiDAR braking guarantees zero collisions.

BENEFITS

Conflict Free Task Consensus

Decentralized auction bidding and version-tracked updates ensure all jobs are settled fairly without duplicate assignments or split-brain conflicts.

Dynamic Hazard Clearance

Computer vision classifies obstructions for efficient rerouting.



RESEARCH AND REFERENCES



SMART INDIA
HACKATHON
2026

1. Peer-Reviewed Academic Research (2021–2026)

- Decentralized Task Allocation (CBBA):S. P. Le et al., IEEE T-RO, 2022 (DOI: 10.1109/TRO.2021.3128410)
- Space-Time Planning (Continuous WHCA* / MAPF):H. Zhang et al., Artificial Intelligence (AIJ), 2022 (DOI: 10.1016/j.artint.2022.103774)
- Reciprocal Collision Avoidance (Kinematic ORCA):M. Alonso-Mora et al., IEEE RA-L, 2021 (DOI: 10.1109/LRA.2021.3062325)
- Multi-Agent Reinforcement Learning (MAPPO):C. Yu et al., NeurIPS, Vol. 35, 2022 (arXiv : 2103.01955)
- Dynamic Travel-Time (ETA) & Queue Regressors : <https://www.mdpi.com/2227-7390/11/7/1723>

2. Industrial Standards & Safety Frameworks

- Industrial Driverless Safety : ISO 3691-4:2023 (Braking zones, hardware veto & personnel detection)
- Decentralized Wireless Mesh Standard : IEEE 802.11s-2021 (Layer-2 B.A.T.M.A.N.-adv mobile fleet relays)

Decentralized middleware: OMG DDS v1.4 / Eclipse Zenoh 1.0. Current deployment uses router-assisted discovery with Zenoh.

3. Modern Open-Source Ecosystem (LTS)

- Core Robotics & Sim : ROS 2 Jazzy Jalisco (LTS 2024–2029) Gazebo Harmonic (LTS 2023–2028)
- Edge Hardware & Vision : Clearpath TurtleBot 4 / Nav2 Stack
- Ultralytics YOLOv8 (2023)

4. Visual Layout & Industry Media Attributions

- MIT News (March 2026):"AI system keeps warehouse robot traffic running smoothly" — news.mit.edu/2026/ai-system-keeps-warehouse-robot-traffic-running-smoothly-0326
- Warehouse Automation Reference Architecture :“ The Role of Autonomous Mobile Robots in Warehouse Automation” — roboticwarehousepicker.com